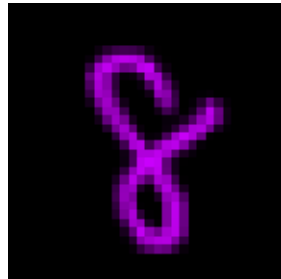
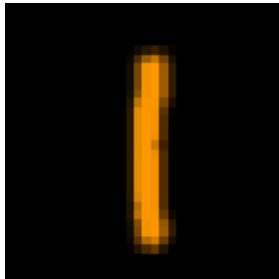
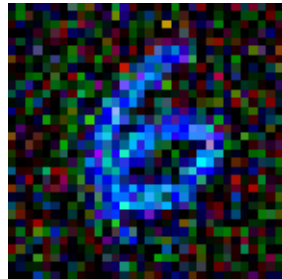
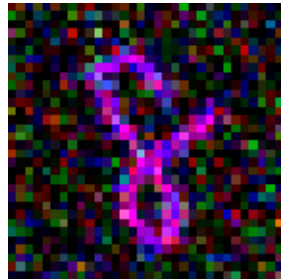
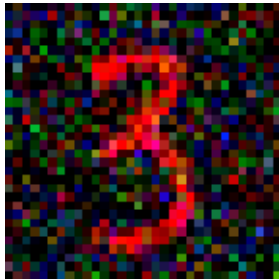
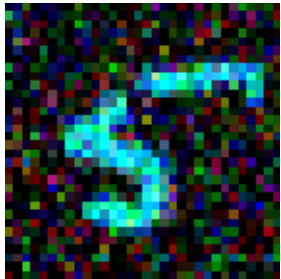
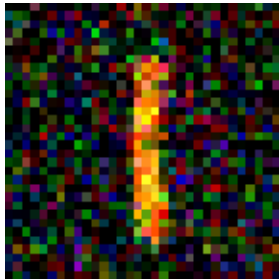


Huber Noise (alpha=1.0, scale=0.5)

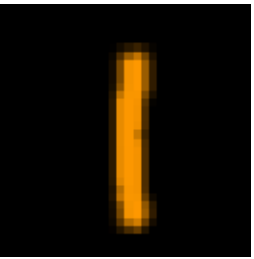
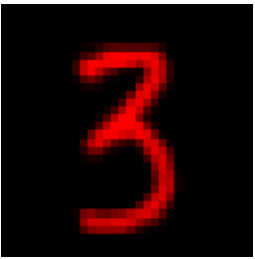
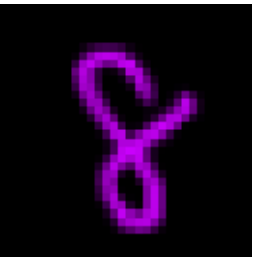
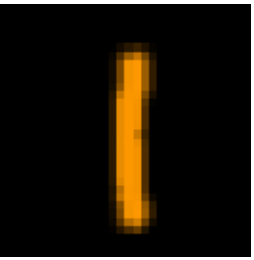
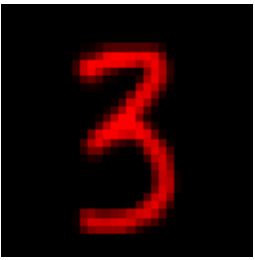
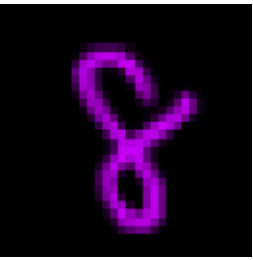
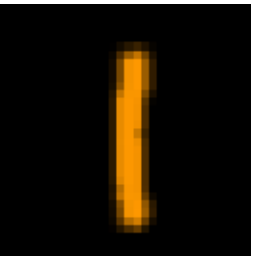
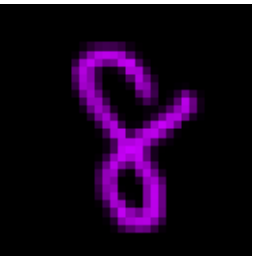
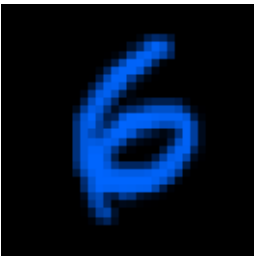
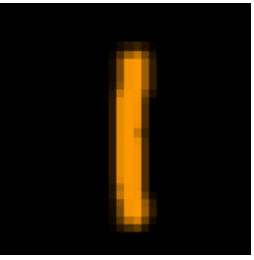
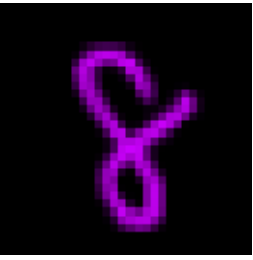
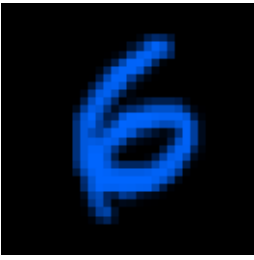
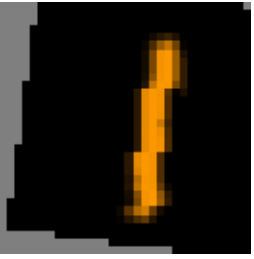
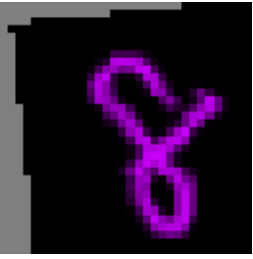
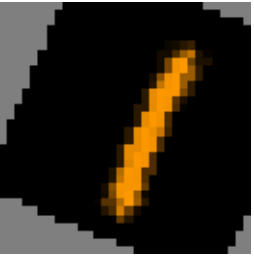
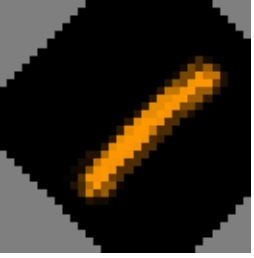
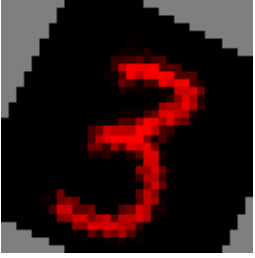
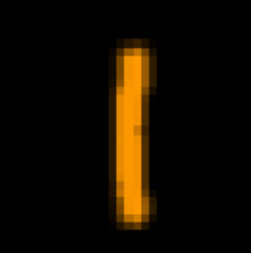
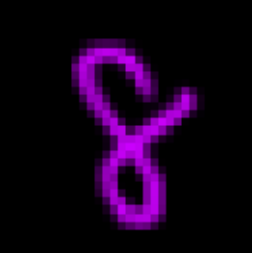
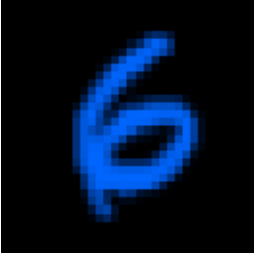
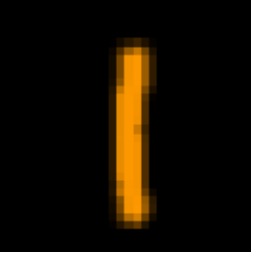
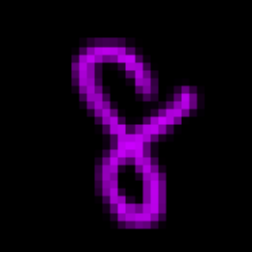
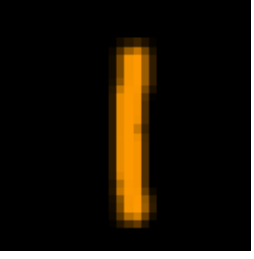
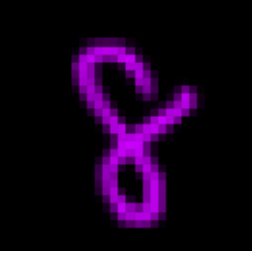
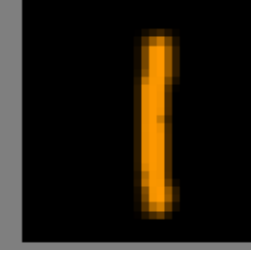
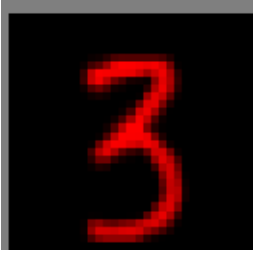
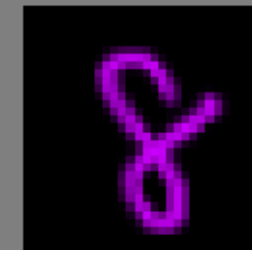
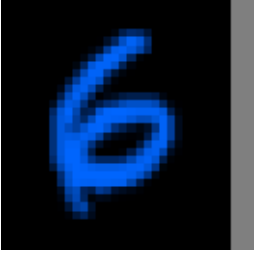
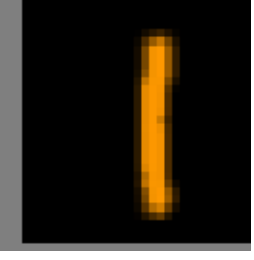
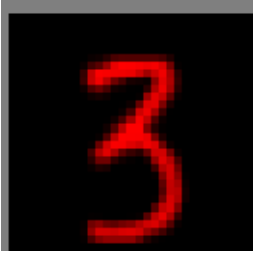
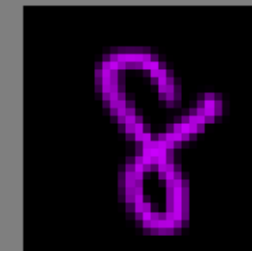
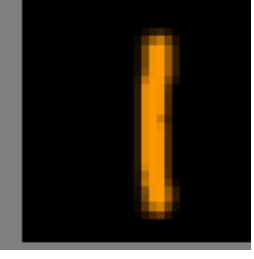
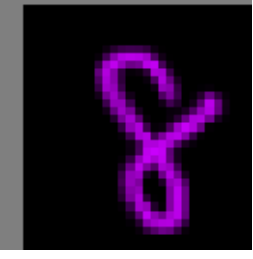
Original



Huber



Camera Shifts Visualization

Original					
Angle (low)					
Angle (med)					
Angle (high)					
Rotation (low)					
Rotation (med)					
Rotation (high)					
Blur (low)					
Blur (med)					
Blur (high)					
Zoom (low)					
Zoom (med)					
Zoom (high)					

Huber Noise - Relative L2 (%)

Method	Alpha=0.0	Alpha=1.0
Abs-LiNGAM (Noisy)	6637.66 ± 1036.08 *	7464.33 ± 1043.72 *
Abs-LiNGAM (Perfect)	107569.20 ± 14574.22 *	115434.52 ± 14671.94 *
BaryCA	3724666.76 ± 466375.25 *	3769651.38 ± 464679.99 *
DiRoCA (eps_20_delta_0.0)	** 427.53 ± 246.36 **	** 505.06 ± 247.21 **
DiRoCA (eps_40_delta_0.0)	1330.93 ± 428.79 *	1604.99 ± 421.62 *
DiRoCA (eps_61.686502153383906_delta_0.0)	3672.23 ± 1100.62 *	4110.02 ± 1086.09 *
DiRoCA (eps_8.0_delta_0.0)	799.27 ± 492.89	894.93 ± 492.68
GradCA	5001759.36 ± 344572.01 *	5065557.24 ± 351951.22 *

** Best Method | * Significantly worse (p<0.05) than best

Camera Shift: Angle - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		6637.66 ± 1036.08 *	6637.66 ± 1036.08 *	6637.66 ± 1036.08 *
Abs-LiNGAM (Perfect)		107569.20 ± 14574.22 *	107569.20 ± 14574.22 *	107569.20 ± 14574.22 *
BaryCA		3724666.76 ± 466375.25 *	3724666.76 ± 466375.25 *	3724666.76 ± 466375.25 *
DiRoCA	(eps_20_delta_0.0)	** 427.53 ± 246.36 **	** 427.53 ± 246.36 **	** 427.53 ± 246.36 **
	(eps_40_delta_0.0)	1330.93 ± 428.79 *	1330.93 ± 428.79 *	1330.93 ± 428.79 *
	(eps_61.686502153383906_delta_0.0)	3672.23 ± 1100.62 *	3672.23 ± 1100.62 *	3672.23 ± 1100.62 *
	(eps_8.0_delta_0.0)	799.27 ± 492.89	799.27 ± 492.89	799.27 ± 492.89
	GradCA	5001759.36 ± 344572.01 *	5001759.36 ± 344572.01 *	5001759.36 ± 344572.01 *

Camera Shift: Rotation - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		4565.70 ± 641.25 *	4052.39 ± 598.97 *	3701.27 ± 576.66 *
Abs-LiNGAM (Perfect)		63305.14 ± 6002.80 *	49993.05 ± 4788.97 *	41072.21 ± 4749.41 *
BaryCA		2139652.65 ± 223831.48 *	1513546.43 ± 161824.25 *	1100777.65 ± 123300.82 *
DiRoCA	(eps_20_delta_0.0)	** 405.39 ± 240.21 **	** 393.97 ± 247.09 **	** 381.52 ± 252.38 **
	(eps_40_delta_0.0)	1090.58 ± 299.99 *	1008.96 ± 295.71 *	939.99 ± 288.15 *
	(eps_61.686502153383906_delta_0.0)	2671.98 ± 688.42 *	2371.94 ± 621.79 *	2129.97 ± 537.41 *
	(eps_8.0_delta_0.0)	785.76 ± 415.32	773.78 ± 396.45	773.17 ± 371.61
	GradCA	2438927.13 ± 175336.12 *	1829350.98 ± 149848.27 *	1381143.24 ± 116758.10 *

Camera Shift: Blur - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		6637.66 ± 1036.08 *	6637.66 ± 1036.08 *	6637.66 ± 1036.08 *
Abs-LiNGAM (Perfect)		107569.20 ± 14574.22 *	107569.20 ± 14574.22 *	107569.20 ± 14574.22 *
BaryCA		3724666.76 ± 466375.25 *	3724666.76 ± 466375.25 *	3724666.76 ± 466375.25 *
DiRoCA	(eps_20_delta_0.0)	** 427.53 ± 246.36 **	** 427.53 ± 246.36 **	** 427.53 ± 246.36 **
	(eps_40_delta_0.0)	1330.93 ± 428.79 *	1330.93 ± 428.79 *	1330.93 ± 428.79 *
	(eps_61.686502153383906_delta_0.0)	3672.23 ± 1100.62 *	3672.23 ± 1100.62 *	3672.23 ± 1100.62 *
	(eps_8.0_delta_0.0)	799.27 ± 492.89	799.27 ± 492.89	799.27 ± 492.89
	GradCA	5001759.36 ± 344572.01 *	5001759.36 ± 344572.01 *	5001759.36 ± 344572.01 *

Camera Shift: Zoom - Relative L2 (%)

DiRoCA	Method	Low	Med	High
	Abs-LiNGAM (Noisy)	4954.00 ± 632.03 *	4920.75 ± 626.59 *	4908.53 ± 638.11 *
	Abs-LiNGAM (Perfect)	71643.10 ± 6240.06 *	71440.30 ± 5862.92 *	71282.14 ± 6426.30 *
	BaryCA	2342652.76 ± 232906.87 *	2340562.64 ± 237009.56 *	2328097.57 ± 224868.20 *
	DiRoCA (eps_20_delta_0.0)	** 410.70 ± 242.21 **	** 409.40 ± 241.51 **	** 410.93 ± 243.70 **
	DiRoCA (eps_40_delta_0.0)	1111.03 ± 300.14 *	1108.27 ± 297.59 *	1111.96 ± 298.86 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	2773.79 ± 711.09 *	2767.03 ± 701.50 *	2763.98 ± 704.20 *
	DiRoCA (eps_8.0_delta_0.0)	794.03 ± 421.42	795.82 ± 420.37	792.54 ± 417.47
	GradCA	2644937.88 ± 183779.07 *	2648016.88 ± 176510.05 *	2646991.79 ± 199924.09 *